Vectors and Linear Algebra

28 October 2026 09:57

Field($\mathbb{F}$) :- Any N.s $\rightarrow \mathbb{F}$ $R = \{-\infty \cdots$

Field addition (+)

① $a, 0 \in \mathbb{F}$

$a + 0 = 0 + a = a$ Additive Identity

② $a, b \in \mathbb{F} \Rightarrow a+b \in \mathbb{F}$

③ $a + (-a) = 0$

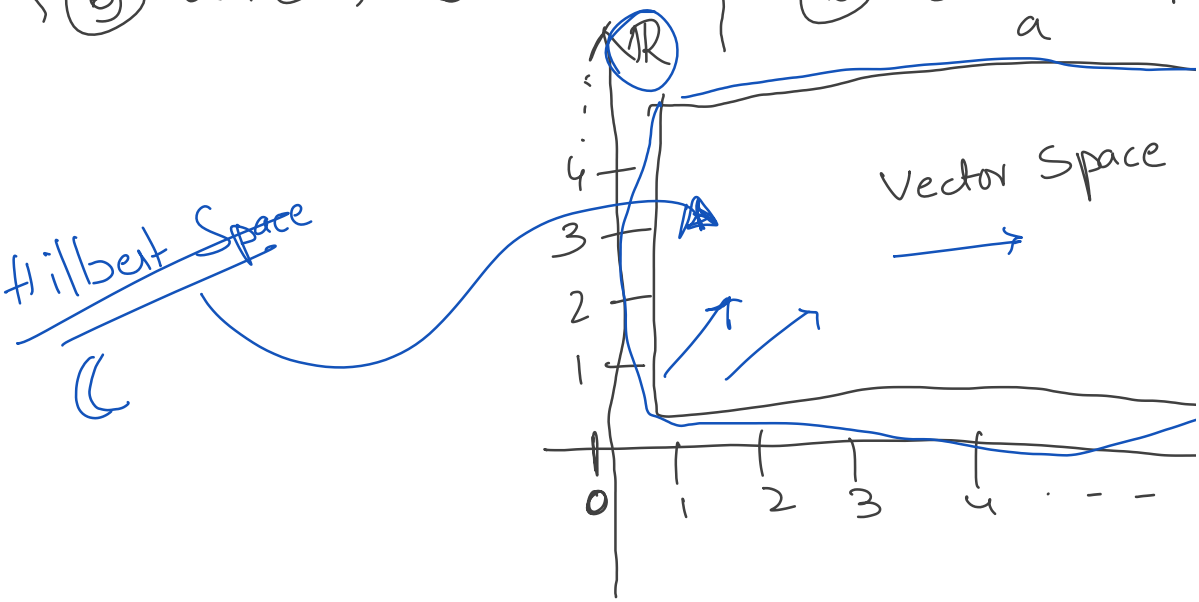
Field Mult

② $a, 1 \in \mathbb{F}$

$a \cdot 1 = 1 \cdot$

④ $a \cdot b \in F$

⑥ $a \cdot \frac{1}{a} = 1$



$a + ib$

↑ Real



$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ 0th / 1st

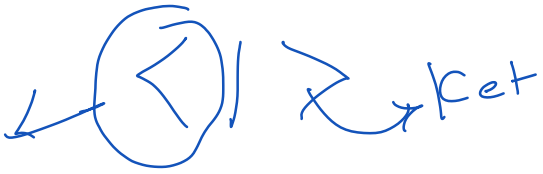
$|1\rangle$ =

ℂ

Dirac Notations ⟶ Bra

orthogonal vectors

bra



$\langle\psi_1|\psi_2\rangle = 0$ ⟶ ψ_1, ψ_2 are or

to

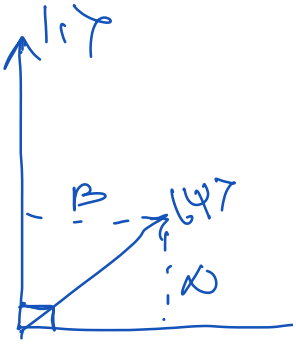
Bra-ket ⟶ Inner / dot pr

$|0\rangle \otimes |1\rangle$

$\langle 0| \cdot |1\rangle$

$\langle 0|1\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix}_{1\times 2} \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{2\times 1} =$

$= 1\cdot 0 + 0\cdot 1 = [0]_{1\times 1}$



$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $X = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

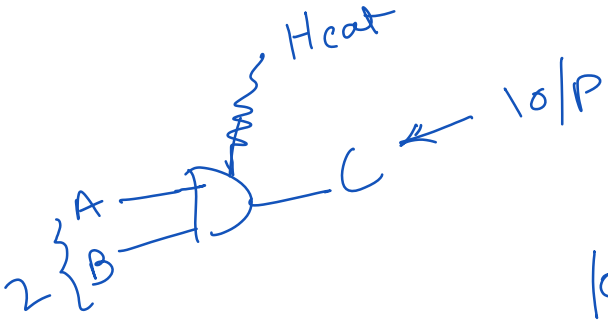
$X|0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$



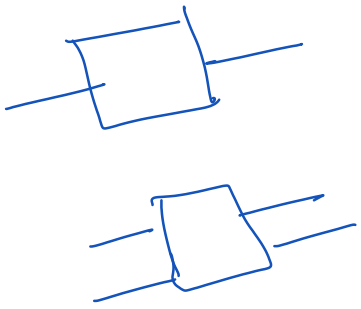
$= 0 \cdot 1$

$1 \cdot 1$

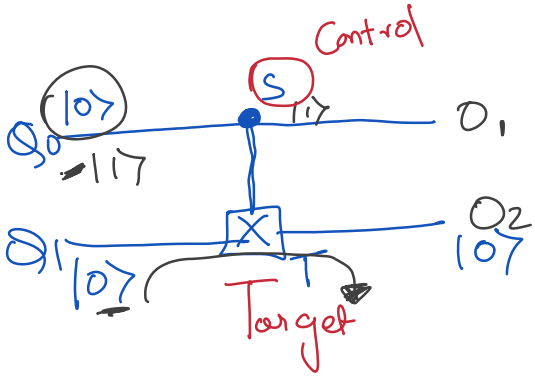
$X|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

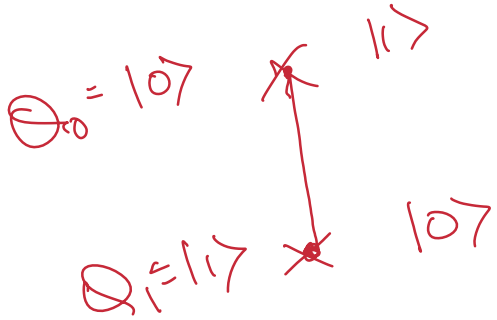


$$\frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{\sqrt{2}\sqrt{2}} = \begin{bmatrix} 1 \\ \end{bmatrix}$$



$$\frac{|00\rangle + |11\rangle}{\sqrt{2}} =$$





C→X ⇒ Controlled X

CCX ⇒ Controlled
Controlle